

Keywords: agentic reinforcement learning • policy optimization • value functions • LLM post-training

SAPO: Single-Rollout Autoregressive Policy Optimization for Agentic RL

A Shared Autoregressive Backbone Yields Policy and Value Functions at No Extra Memory Cost, Beating PPO and GRPO on Multi-Turn Agent Tasks

By Dayang Liang, Lang Feng, Bo An, Yunlong Liu

arXiv:2608.19842 • Preprint, August 2026

Reinforcement learning for multi-turn LLM agents faces a harsh trade-off: critic-based methods like PPO provide token-level credit but double memory overhead, while critic-free methods like GRPO avoid a separate model but assign reward only at sequence level and stall on straggler rollouts. SAPO escapes the dilemma by reading policy distributions and value estimates from *the same* autoregressive backbone at distinct causal boundaries, then optimizing both with a single rollout per step. On ALF-World and WebShop, SAPO outperforms PPO by +15.1 pp and GRPO by +12.1 pp while cutting per-iteration runtime by 33.2%.

AT A GLANCE

Problem: Agentic RL for LLMs requires either a costly separate critic (PPO) or coarse sequence-level credit from large rollout groups (GRPO), with neither efficiently handling long-horizon tasks.

Why it matters: LLM agent deployment is blocked by training cost; a single-rollout method that matches critic-based credit quality would substantially lower the barrier to scaling.

Method: Exploit causal boundaries in the autoregressive backbone to produce policy and value heads from shared parameters; optimize PPO objectives jointly with auxiliary on-policy SARSA for value heads.

Result: +15.1 pp over PPO and +12.1 pp over GRPO on ALFWorld/WebShop with Qwen2.5-1.5B/7B; 33.2% faster per iteration.

Evidence: Ablations show that batch-normalized GAE is critical for stability; removing either the SARSA auxiliary or the shared backbone degrades performance substantially.

Agentic reinforcement learning trains LLMs to complete multi-turn tasks — web navigation, code execution, tool use — by interacting with an environment over many steps. Reward is sparse and delayed: the agent may execute dozens of actions before receiving a signal about whether the task succeeded.

Two RL paradigms dominate this setting. **PPO** maintains a separate critic model V_ϕ that predicts future return from each intermediate state, enabling fine-grained *token-level* advantage estimates. But the critic roughly doubles memory footprint and adds a second forward-backward pass per training step. **GRPO** eliminates the critic by sampling G rollouts per prompt and using group statistics as a baseline, but credit is assigned only at sequence level, and throughput is bottlenecked by the slowest rollout in each group.

SAPO proposes a third path: derive both the policy distribution and value estimate from a *single* autoregressive forward pass over a single rollout, sharing all parameters while keeping the two estimates informationally independent.

Why Existing Approaches Fall Short

PPO: The critic doubles memory, complicating deployment on commodity hardware. Training the critic on the policy’s own token context is non-stationary — as the policy changes, the value targets shift — causing instability that practitioners address with separate learning rates, KL penalties, and warm-up schedules.

GRPO and group-relative methods: Sequence-level credit means all tokens in a response receive the same advantage signal, even though early tokens set up the trajectory and late tokens close it. This masks the contribution of individual actions, slowing learning on tasks with long chains of dependencies. The straggler problem compounds this: when $G = 8$, training stalls until all eight rollouts finish, wasting compute proportional to the gap between fastest and slowest.

Prior shared-backbone attempts have fused policy and value heads at the final layer but computed them over the same token context. This conflates the *action* decision (next token) with the *state value* prediction (return from here), conflating two informationally distinct quantities.

Core Mechanism

SAPO exploits the *causal boundary* property of autoregressive transformers. At step t in the agent’s trajectory, the residual stream at position j contains information from all tokens at positions $\leq j$ but not from positions $> j$. This means:

The Big Picture

- The **policy head**, reading the stream at the position of the last token of the current observation, sees the full observation context but no generated action tokens.
- The **value head**, reading at the position just after the first action token is generated, sees the observation and the first action token — a distinct information set.

These two heads share *all parameters* (the transformer backbone), but they are causally separated: neither head’s input is derived from the other’s output position. This makes them informationally independent despite sharing weights.

Technical Formulation

Let s_t be the observation (token sequence) at step t and a_t the action generated by the policy π_θ . The shared backbone f_θ produces logits at two causal boundaries:

$$\pi_\theta(a_t|s_t) = \text{softmax}(W_\pi \cdot f_\theta(s_t)[j_\pi])$$

$$V_\theta(s_t) = W_V \cdot f_\theta(s_t \| a_t^{(0)})[j_V]$$

where j_π is the position of the last observation token and j_V is the position after the first generated action token, and $\|$ denotes concatenation.

The total loss combines the clipped PPO objective and an auxiliary on-policy SARSA value loss:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}}(\pi_\theta) + \lambda \mathcal{L}_V(\theta)$$

$$\mathcal{L}_{\text{PPO}} = -\mathbb{E}_t \left[\min \left(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right]$$

$$\mathcal{L}_V = \mathbb{E}_t \left[(V_\theta(s_t) - r_t - \gamma V_{\theta_{\text{old}}}(s_{t+1}))^2 \right]$$

where $\rho_t = \pi_\theta(a_t|s_t) / \pi_{\theta_{\text{old}}}(a_t|s_t)$.

Advantages use a batch-normalized λ -return GAE:

$$\hat{A}_t = \frac{\delta_t^\lambda - \mu_B}{\sigma_B + \varepsilon}, \quad \delta_t^\lambda = \sum_{k=0}^{T-t} (\gamma \lambda)^k (r_{t+k} + \gamma V_\theta(s_{t+k+1}) - V_\theta(s_{t+k}))$$

where μ_B, σ_B are the batch mean and standard deviation of unnormalized advantage estimates.

Learning or Inference Procedure

Each training iteration proceeds as:

1. Sample one rollout per prompt from the current policy π_θ .
2. Compute value estimates at all causal boundaries using the shared backbone.
3. Compute λ -return advantages and apply batch normalization.
4. Perform K gradient steps on $\mathcal{L}$ with the PPO probability-ratio clip preventing large policy updates.

5. Update $\theta_{\text{old}} \leftarrow \theta$.

At inference time, only the policy head is used; value heads are discarded. No rollout groups, no separate critic model.

What the Guarantee Says

No formal convergence guarantee is provided. The theoretical contribution is the causal-boundary argument establishing that the two heads are informationally independent despite sharing parameters. The empirical claim is that this independence is sufficient to stabilize joint optimization of the PPO and SARSA objectives.

Experimental Findings

- **ALFWorld success rate:** SAPO (Qwen2.5-7B) achieves 78.4% vs. PPO 62.1% and GRPO 65.3%.
- **WebShop success rate:** SAPO achieves 54.7% vs. PPO 40.8% and GRPO 43.9%.
- **Average improvement:** +15.1 pp vs. PPO; +12.1 pp vs. GRPO, across model sizes and benchmarks.
- **Runtime:** 33.2% faster per iteration than PPO at matched throughput; comparable to GRPO with $G = 1$ but with much better credit assignment.
- **Memory:** No separate critic model; SAPO’s memory footprint matches an inference-only LLM deployment.

Ablations and Interpretation

Batch normalization of advantages: Removing batch normalization destabilizes training on long trajectories (>20 steps). The distribution of unnormalized advantages shifts as the policy improves, causing the PPO clip to become inactive.

Auxiliary SARSA loss: Removing $\mathcal{L}_V$ degrades performance to near-GRPO levels, confirming that the value heads must be trained, not just read passively.

Causal separation: Ablating to the same position for both heads (reading from the final observation token for both policy and value) reduces performance by 4–6 pp, confirming that informational separation is necessary for independent credit assignment.

Model size: The 1.5B model benefits more from SAPO than the 7B model in absolute terms, suggesting that smaller models have more to gain from precise credit assignment.

Reference

Dayang Liang, Lang Feng, Bo An, Yunlong Liu. **SAPO: Single-Rollout Autoregressive Policy Optimization for Agentic Reinforcement Learning.**

arXiv:2608.19842, August 2026. [https://arxiv.org/
abs/2608.19842](https://arxiv.org/abs/2608.19842)

Keywords: robot foundation model • world model • test-time compute • vision-language-action

τ_0 -VLA: a Hierarchical Robot Foundation Model with World-Model-Guided Test-Time Computation

When the Policy Is Unsure, a World Model Simulates Candidates and a Value Model Picks the Winner—Scaling Test-Time Compute for Long-Horizon Robot Manipulation

By SII Research

arXiv:2608.16885 • Preprint, August 2026

Long-horizon robot manipulation requires executing dozens of subtasks reliably and sequencing them coherently. Most hierarchical vision-language-action (VLA) models make high-level decisions with a single forward pass, leaving no mechanism to allocate more computation to difficult or ambiguous choices. τ_0 -VLA introduces confidence-gated test-time compute: when token-level confidence falls below a threshold, the model triggers a propose–predict–evaluate loop in which a world model simulates outcomes and a value model selects the best subtask. Trained on 40,115 hours of real-world manipulation data, τ_0 -VLA achieves +14.2 pp in long-horizon success rate over its single-pass baseline.

AT A GLANCE

Problem: Hierarchical VLA models commit to subtask decisions in one forward pass with no mechanism to spend more compute on hard decisions, causing failures on ambiguous or novel long-horizon tasks.

Why it matters: Long-horizon manipulation is the frontier of robot capability; scaling test-time compute for planning, not just reasoning, could unlock qualitatively new task categories.

Method: Monitor token-confidence of the high-level policy; when below threshold, invoke a world-model propose–predict–evaluate loop to score candidate subtasks by predicted outcome quality.

Result: +14.2 pp closed-loop success on long-horizon tasks; +9.8 pp subtask prediction accuracy with 8 candidates vs. 1.

Evidence: World model prediction quality correlates with value model accuracy ($r = 0.81$); ablations confirm the loop only activates for genuinely hard decisions (78% precision).

Hierarchical robot control separates the problem of *what to do* (the high-level policy, which generates subtask goals like “pick up the cup” or “open the drawer”) from *how to do it* (the low-level skill executor, which produces motor commands). This separation is powerful: the high-level policy can reason about task semantics, while the low-level policy handles motor precision.

The bottleneck is the high-level policy’s decision quality. On a familiar task, a single forward pass is sufficient. But on a novel combination of objects, or after an unexpected intermediate state caused by imprecise skill execution, the high-level policy may be genuinely uncertain about the right next subtask. A wrong high-level decision cascades through all subsequent skill execution and can be unrecoverable.

τ_0 -VLA makes high-level planning compute-scalable by adapting the number of forward passes to the confidence of the decision.

Why Existing Approaches Fall Short

Single-pass hierarchical VLAs (e.g., SayCan, UniFS) generate subtask goals with one forward pass through a vision-language model. This is fast but brittle: the model either outputs the correct subtask or it does not, with no opportunity to explore alternatives.

Tree search methods (e.g., Monte Carlo Tree Search applied to robot planning) require executing candidate subtasks in a real or simulated environment to estimate their value, which is prohibitively expensive for physical manipulation tasks.

LLM-based replanning repeats the forward pass with rephrased prompts when a subtask fails, but this is reactive rather than proactive: it corrects after failure rather than anticipating it. It also requires a binary success/failure signal, which may not be available for subtasks that partially succeed.

τ_0 -VLA’s key contribution is *proactive* uncertainty detection plus *simulated* candidate evaluation — avoiding the cost of physical trial-and-error.

Core Mechanism

Three modules work together:

Memory-augmented high-level policy π_{high} : A transformer that reads the current RGB observation o_t and an execution memory m_t (a compressed representation of past subtasks and their outcomes), and generates subtask s_t autoregressively.

Confidence monitor: After generating s_t , compute the mean log-probability of the generated tokens as a

The Big Picture

confidence score:

$$c_t = \exp \left(\frac{1}{|s_t|} \sum_{i=1}^{|s_t|} \log \pi_{\text{high}}(s_t^{(i)} | o_t, m_t, s_t^{(<i)}) \right)$$

If $c_t \geq \tau_c$, commit to s_t . If $c_t < \tau_c$, trigger the extended loop.

Propose–predict–evaluate loop: Generate K candidate subtasks via nucleus sampling, predict the terminal observation induced by each, and score with a value model:

$$s^* = \arg \max_{k \in [K]} V(s^k, \hat{o}_{t+1}^k, m_t)$$

Technical Formulation

Let the world model $\mathcal{W}_\psi$ predict the terminal RGB observation after executing subtask s^k from state (o_t, m_t) :

$$\hat{o}_{t+1}^k = \mathcal{W}_\psi(o_t, m_t, s^k)$$

The value model V_ν estimates task progress from the predicted terminal state and memory:

$$V_\nu(s^k, \hat{o}_{t+1}^k, m_t) \in [0, 1]$$

representing the estimated probability of ultimate task success given this subtask choice.

The selected subtask is:

$$s^* = \arg \max_{k=1}^K V_\nu(s^k, \hat{o}_{t+1}^k, m_t)$$

The execution memory update after executing s^* :

$$m_{t+1} = \text{GRU}_\omega(m_t, [s^*; \text{success}_t])$$

where $\text{success}_t \in \{0, 1\}$ is the low-level success signal and GRU_ω is a gated recurrent unit that compresses history into a fixed-size memory vector.

Learning or Inference Procedure

High-level policy training: Supervised next-subtask prediction over real demonstration trajectories with cross-entropy loss. No RL is used for the high-level policy.

World model training: Diffusion-based next-observation prediction conditioned on (o_t, m_t, s^k) . Trained on (o_t, s_t, o_{t+1}) triples extracted from demonstrations.

Value model training: Trained on $(o_t, m_t, s_t, \text{task_success})$ tuples where task_success is the binary outcome of the full trajectory. Positive examples have the correct subtask; negatives use counterfactual subtasks sampled from the policy distribution.

Data: 40,115 hours of real-world manipulation data across tabletop, kitchen, and dexterous manipulation domains.

What the Guarantee Says

No formal guarantees are provided. The key empirical claim is that the confidence score c_t is well-calibrated: low-confidence decisions are harder (lower subtask accuracy at $c_t < \tau_c$ vs. $c_t \geq \tau_c$), and additional compute on low-confidence steps provides the largest gains.

Experimental Findings

- **Long-horizon closed-loop success:** +14.2 pp over single-pass policy, averaged across 6 task categories.
- **Subtask prediction accuracy:** +9.8 pp improvement by increasing K from 1 to 8 candidates on held-out tasks.
- **Confidence calibration:** Low-confidence decisions ($c_t < \tau_c$) have 31% lower single-pass subtask accuracy than high-confidence decisions, validating the trigger.
- **World model quality:** Spearman correlation $r = 0.81$ between world model prediction quality and value model accuracy, validating the end-to-end pipeline.
- **Compute efficiency:** The propose–predict–evaluate loop triggers on $\sim 22\%$ of steps, keeping average inference overhead below $2.4\times$ the single-pass baseline.

Ablations and Interpretation

Confidence threshold τ_c : Lower thresholds trigger the loop more frequently, improving accuracy but increasing latency. The optimal $\tau_c = 0.72$ triggers on 22% of steps for best cost-accuracy tradeoff.

Number of candidates K : Gains diminish beyond $K = 8$; marginal improvement from $K = 4$ to $K = 8$ is less than half the improvement from $K = 1$ to $K = 4$.

World model vs. no prediction: Selecting the highest policy-probability candidate (no world model) without predicting outcomes achieves only +3.1 pp over the baseline, vs. +14.2 pp with world-model-scored selection.

Memory ablation: Replacing the GRU memory with no-context (Markov) decision making reduces long-horizon success by 11.4 pp, confirming that history compression is essential.

Reference

SII Research. **τ_0 -VLA: a Hierarchical Robot Foundation Model with World-Model-Guided Test-Time Computation.** arXiv:2608.16885, August 2026. <https://arxiv.org/abs/2608.16885>

Keywords: LLM reinforcement learning • credit assignment
• value functions • privileged information

Le Critique: Privileged Value Functions for LLM Reinforcement Learning

Critics with Access to Task-Privileged Context Provide Token-Level Advantages Without Contaminating the Policy, Beating GRPO with Fewer Rollouts

By Siddarth Venkatraman, Matthieu Dinot, Laurence Aitchison

arXiv:2608.16739 • Preprint, August 2026

Group-relative methods like GRPO reduce gradient variance by sampling multiple rollouts per prompt, but they assign reward only at sequence level and are bottlenecked by straggler rollouts that reduce throughput and inflate off-policyyness. Learned value functions solve both problems in principle, but training an accurate critic from the policy’s own context is notoriously hard. *Le Critique* introduces Privileged Value Functions (PVF), which give the critic access to task-privileged information (ground-truth answer, rubric, execution state) unavailable to the policy at inference time, enabling precise token-level advantage estimates from a single rollout. An adaptive interpolation baseline (TETHER) gracefully degrades to group-relative methods when the critic is unreliable.

AT A GLANCE

Problem: GRPO provides only sequence-level credit and suffers from straggler bottlenecks; standard value functions are hard to train from the policy’s own context.

Why it matters: Token-level credit assignment could dramatically accelerate LLM RL by identifying which reasoning steps actually contribute to task success.

Method: Privileged Value Functions give critics access to task-privileged context (e.g., the ground-truth) during training only; TETHER adaptively interpolates between PVF and group baselines depending on estimated critic reliability.

Result: +4.2 accuracy points on GSM8K/MATH vs. GRPO at the same training budget; $2.3\times$ faster convergence than PPO.

Evidence: Ablations confirm that privileged context is essential; PVF without privileged information underperforms GRPO, while TETHER provides robustness when critic quality varies.

The Big Picture

Reinforcement learning for LLMs improves model behavior on tasks with scalar reward: math problem solving, code generation, instruction following. The central challenge is efficient credit assignment: which tokens in a response contributed to earning the reward, and by how much?

Credit assignment is easy when reward is dense (every token gets a signal) or when rollouts are very short (a one-step response). For multi-step reasoning, answer construction, or long tool-use sequences, credit is sparse and long-delayed. The community has largely converged on two approaches, each with a critical flaw.

GRPO avoids a critic by constructing a group of G rollouts and using group statistics as a baseline. The flaw: every token in a response receives the same advantage, regardless of whether it was a key reasoning step or filler text.

PPO with a critic provides token-level advantages but requires training a separate model V_ϕ that can accurately predict return from the policy’s token context. The flaw: the context available to the critic (the tokens generated so far) is the same context available to the policy — and the policy’s context does not contain enough information to reliably predict long-horizon return.

Le Critique’s insight: the critic does not need to be limited to what the *policy* can see.

Why Existing Approaches Fall Short

Sequence-level baselines (GRPO, REINFORCE++): Assigning the same advantage to all tokens in a 200-token reasoning chain means the model cannot distinguish a critical deduction that led to the correct answer from a redundant step that happened to precede it. On math reasoning tasks, early reasoning steps (problem setup, variable definition) are causally most important, but they receive the same credit as conclusion steps under group-relative methods.

Standard value functions (PPO): The policy’s context at step t contains the problem statement and all tokens generated so far. This context does not directly contain the ground-truth answer, the rubric used for scoring, or whether the model is “on track” toward the correct solution. Trained on this incomplete information, critics tend to be poorly calibrated: they predict high value for high-confidence-but-wrong responses and fail to distinguish correct from incorrect intermediate reasoning.

Core Mechanism

Privileged Value Functions (PVF): The critic $V_\phi(s_t, p_t)$ receives both the standard policy context s_t (the token sequence) and a privileged context p_t available

in the training environment but not to the policy during inference. For math reasoning, p_t might include the ground-truth answer and intermediate solution steps; for code generation, it might include the test case outputs.

The privileged context is concatenated before the model’s context window at the critic input level and is never passed through the policy’s forward pass. The policy gradient uses advantages derived from the PVF:

$$\hat{A}_t = r_t + \gamma V_\phi(s_{t+1}, p_{t+1}) - V_\phi(s_t, p_t)$$

The policy parameters θ are updated via:

$$\nabla_\theta J = \mathbb{E}_t[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot \hat{A}_t]$$

Gradients through the critic are blocked: ϕ is updated separately with mean squared TD error.

TETHER: Not all critics are equally reliable. TETHER adaptively interpolates between the PVF baseline and the group-relative baseline:

$$b_t = \hat{\rho} \cdot V_\phi(s_t, p_t) + (1 - \hat{\rho}) \cdot \mathbb{E}_G[R(\mathcal{T})]$$

where $\hat{\rho} \in [0, 1]$ estimates critic reliability from the held-out validation loss:

$$\hat{\rho} = \sigma(-\alpha \cdot \mathcal{L}_V^{\text{hold-out}})$$

High $\mathcal{L}_V$ (unreliable critic) drives $\hat{\rho} \rightarrow 0$, recovering group-relative training; low $\mathcal{L}_V$ (reliable critic) drives $\hat{\rho} \rightarrow 1$, using token-level PVF advantages.

Technical Formulation

Let π_θ be the LLM policy and V_ϕ the PVF critic. During a training step, one rollout $\mathcal{T} = (s_0, a_0, \dots, s_T, r_T)$ is generated from π_θ . The full TETHER policy gradient is:

$$\nabla_\theta J = \mathbb{E}_\mathcal{T} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot \hat{A}_t^{\text{TETHER}} \right]$$

$$\hat{A}_t^{\text{TETHER}} = r_t + \gamma b_{t+1} - b_t$$

The critic is updated independently with:

$$\mathcal{L}_V = \mathbb{E}_t[(V_\phi(s_t, p_t) - (r_t + \gamma V_\phi(s_{t+1}, p_{t+1})))^2]$$

Learning or Inference Procedure

Training: Privileged context p_t is constructed from ground-truth task data (available in supervised datasets) and is passed to the critic at each step. The critic is updated every training iteration; reliability $\hat{\rho}$ is recomputed from a 10% held-out portion of each batch.

Inference: The critic and its privileged context are discarded entirely. The policy π_θ runs without any modifications, and privileged information is never exposed during deployment.

What the Guarantee Says

No formal guarantee is provided for PVF. The key theoretical observation is that the privileged critic, having access to the ground truth, has a much lower Bayes-optimal value estimation error than a critic trained from the policy’s context alone. This tightened value estimation translates to lower-variance advantage estimates, which in turn reduces the sample complexity of policy improvement.

Experimental Findings

- **Math reasoning (GSM8K + MATH):** PVF with TETHER achieves +4.2 accuracy points vs. GRPO at the same training budget (number of rollouts and gradient steps).
- **Training efficiency:** TETHER reaches GRPO’s final accuracy 2.3× faster in training steps, using fewer rollouts per gradient step.
- **Straggler reduction:** Eliminating the rollout-group requirement reduces training wall-clock time by 18–24% on 8-GPU configurations where straggler variance is highest.
- **PVF vs. standard critic:** PVF outperforms a standard PPO critic by +2.9 accuracy points, confirming that privileged context is the source of the advantage, not just having a critic.

Ablations and Interpretation

Privileged context content: Ground-truth answer provides most of the benefit; adding intermediate solution steps provides marginal additional gain. This suggests that knowing the target outcome is the most valuable privileged information.

TETHER vs. fixed interpolation: Fixed $\hat{\rho} = 0.5$ performs 1.8 pp worse than adaptive TETHER, confirming that critic reliability varies during training and adaptivity is needed.

Rollout group size: TETHER with $G = 1$ (single rollout) outperforms GRPO with $G = 8$ by 1.2 accuracy points, demonstrating that privileged credit assignment compensates for reduced group size.

Offline application: PVF-trained policies retain their advantage over GRPO baselines on out-of-distribution test prompts, suggesting that the improved credit assignment produces a qualitatively better-learned policy, not just in-distribution fitting.

Reference

Siddarth Venkatraman, Matthieu Dinot, Laurence Aitchison. **Le Critique: Privileged Value Functions for LLM Reinforcement Learning.** arXiv:2608.16739, Au-

gust 2026. <https://arxiv.org/abs/2608.16739>

Keywords: vision-language models • benchmark • visual decision making • constraint satisfaction

ComplexityWorld: Benchmarking Vision-Language Models on Verifiable Visual Decision Making

390 Tasks Across 39 Visual Worlds Show That Most VLMs Cannot Extract Constraints from Images and Satisfy Them Jointly — Only One Model Exceeds 75%

By ComplexityWorld Authors

arXiv:2608.07584 • Preprint, August 2026

Many real-world AI tasks require more than perceiving what an image contains: a model must use visual evidence to make a decision whose components jointly satisfy global constraints. ComplexityWorld introduces a benchmark of 390 tasks across 39 domain-inspired visual worlds and 29 decision categories, each scored by an executable verifier that accepts any feasible solution. Under direct visual inference, all evaluated VLMs except GPT-5.6-Sol fall below 40% verifier acceptance rate (VAR), while GPT-5.6-Sol reaches 75.6%. A 23-point gap between visual and structured-text conditions reveals that the dominant failure mode is visual constraint extraction, not reasoning.

AT A GLANCE

Problem: VLM benchmarks test perception and QA accuracy but not the ability to make constraint-satisfying decisions from visual evidence, leaving a large deployment-relevant gap unmeasured.

Why it matters: Real-world tasks (scheduling, logistics, design) require visually-grounded constraint satisfaction; current evaluations do not capture whether models can do this.

Method: Generate 390 tasks from hidden structured specifications rendered as visual scenes; use executable verifiers rather than LLM judges to score feasibility.

Result: Only GPT-5.6-Sol exceeds 40% VAR (reaching 75.6%); all other models remain below 40%.

Evidence: Same tasks in structured-text form achieve +23 pp average VAR, isolating visual extraction as the key failure mode.

The Big Picture

Vision-language models have advanced rapidly on perception tasks — they can accurately describe image contents, count objects, and answer factual questions about visual

scenes. This progress has driven optimism about deploying VLMs in real-world decision support: assisting with medical imaging, factory floor monitoring, logistics planning, and scientific data analysis.

These real-world tasks share a property that standard benchmarks do not test: they require making a *decision* from visual evidence, where the decision must satisfy multi-part constraints. A scheduling assistant reviewing a visual calendar cannot merely describe meetings; it must identify conflicts, dependencies, and resource constraints from the visual layout and produce a feasible schedule satisfying all of them simultaneously.

This class of problem — *visual constraint-satisfying decision making* — is distinct from visual perception and from verbal reasoning, and it is largely unmeasured by current evaluation suites. ComplexityWorld is the first systematic benchmark for this capability.

Why Existing Approaches Fall Short

Perception benchmarks (VQA, object counting, scene description) test whether models correctly identify what is present in an image. They do not test whether models can extract structured decision-relevant information from visual representations of constraints.

Verbal reasoning benchmarks (GSM8K, logical reasoning datasets) test constraint satisfaction in text, but cannot assess whether visual extraction precedes and enables correct reasoning.

Prior VLM decision-making benchmarks have used LLM-as-judge evaluation, which is susceptible to hallucination and cannot distinguish a feasible from a plausible-sounding infeasible solution. ComplexityWorld’s verifier-based evaluation avoids these issues.

Core Mechanism

Task generation pipeline: Each task starts from a *hidden structured specification* $\mathcal{S}$ that defines:

- A set of objects, resources, or agents with attributes
- A set of hard constraints over decision variables
- A decision space (assignments, routes, orderings, partitions)

The specification is rendered into a visual scene I using a domain-appropriate renderer (grid-based for scheduling, graph-based for routing, CAD-based for spatial arrangement). An executable verifier $\mathcal{V}$ checks whether any response r satisfies all constraints in $\mathcal{S}$.

Key design choices: The verifier accepts *any* feasible solution, not just the canonical one, avoiding benchmark contamination and penalizing models solely for constraint violations rather than stylistic deviation.

Technical Formulation

Let $\mathcal{S} = (X, C, \mathcal{D})$ be a task specification with decision variables $X = \{x_1, \dots, x_n\}$, constraints $C = \{c_1, \dots, c_m\}$ over X , and domain $\mathcal{D}$.

A response r parsed from the model’s text output is a partial or complete assignment $\hat{X} = \{\hat{x}_1, \dots, \hat{x}_n\}$. Verifier acceptance:

$$\text{Accept}(r, \mathcal{S}) = \mathbf{1} \left[\bigwedge_{j=1}^m c_j(\hat{X}) = \text{TRUE} \right]$$

The verifier acceptance rate (VAR) for model M over task set $\mathcal{T}$:

$$\text{VAR}(M) = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \text{Accept}(M(I_i), \mathcal{S}_i)$$

The visual constraint extraction gap is measured by the difference between structured-text and visual-scene performance:

$$\Delta_{\text{VCE}}(M) = \text{VAR}(M, \text{structured}) - \text{VAR}(M, \text{visual})$$

Learning or Inference Procedure

ComplexityWorld is an evaluation benchmark, not a training procedure. Models are evaluated zero-shot with a standardized prompt that specifies the task domain, output format, and response constraints. No few-shot examples are provided in the main evaluation; ablations with $k = 1, 3, 5$ in-context examples show improvement but do not close the gap with structured-text performance.

What the Guarantee Says

The verifier is formally specified for each domain using executable constraint logic. Verifier correctness is validated by human experts who confirm that feasibility checks match domain semantics. This guarantees that VAR measures genuine constraint satisfaction, not LLM-judge agreement with plausible-sounding responses.

Experimental Findings

- **Direct visual inference VAR:** GPT-5.6-Sol reaches 75.6%; all other models remain below 40%.
- **Visual constraint extraction gap:** Average $\Delta_{\text{VCE}} = +23.1$ pp when the same task is given in structured text vs. as a visual scene.
- **Presentation sensitivity:** Changing visual style (color scheme, font, layout) while keeping constraints identical shifts VAR by up to 18 pp for the same model, revealing brittleness in visual feature extraction.

- **Category variation:** Routing and scheduling domains are hardest (mean VAR < 25%); assignment and partitioning domains are easier (mean VAR ~38%).
- **Open vs. proprietary models:** Best open-weight model achieves 31.4% VAR; best proprietary model (GPT-5.6-Sol) achieves 75.6%, a gap of 44.2 pp.

Ablations and Interpretation

In-context examples: Adding $k = 5$ in-context (image, decision) examples improves VAR by 6.2 pp on average, but narrows the visual-structured gap by only 1.3 pp, confirming that the bottleneck is visual extraction, not few-shot learning of the decision format.

Chain-of-thought prompting: Requiring models to reason step-by-step before producing a decision improves VAR by 4.8 pp on average, with larger gains on routing (+8.1 pp) and scheduling (+7.3 pp) tasks that have natural sequential decision structure.

Constraint complexity: VAR decreases sharply with the number of constraints m : from 52% at $m = 2$ to 21% at $m = 6$ for a representative model, suggesting exponential constraint combinatorics as the dominant difficulty driver.

Error analysis: Manual inspection of 200 failures reveals three failure modes: misreading a constraint value from the image (43%), correctly extracting constraints but producing an infeasible assignment (38%), and producing a feasible partial assignment that misses some variables (19%).

Reference

ComplexityWorld Authors. **ComplexityWorld: Benchmarking Vision-Language Models on Verifiable Visual Decision Making.** arXiv:2608.07584, August 2026. <https://arxiv.org/abs/2608.07584>

Keywords: VLM evaluation • behavioral reliability • scientific figure understanding • confabulation

How Do VLMs Behave When Blind or Misled? Behavioral Evaluation of VLMs on Scientific Figures

SciFigBench Reveals That VLMs Confabulate Data Values When Figures Are Blurred and Defer to Incorrect Captions Over Visual Evidence

By Paul Osemudiamé Oamen, Owusu-Banahene Osei, Ananya Mukherjee, Christian Greisinger, Steffen Eger, Pius Onobhayedo, Wei Zhao

arXiv:2608.13267 • Preprint, August 2026

AI assistants for scientific reading increasingly interpret research figures, but standard benchmarks test only accuracy on clean images. SciFigBench stress-tests VLMs on 250 scientific figures across 34,000+ evaluation setups probing perception, reasoning, and behavioral reliability under uncertainty. The findings are alarming: VLMs confabulate specific data values in 62–84% of cases when critical figure elements are blurred, and defer to incorrect captions over accurate visual evidence 41–67% of the time. Models ranked highest on standard accuracy correlate poorly with behavioral reliability ($\rho = 0.31$), indicating that safe scientific deployment cannot be inferred from existing evaluations.

AT A GLANCE

Problem: Standard VLM benchmarks measure accuracy on clean figures but not behavioral reliability when visual evidence is missing or misleading — a critical gap for scientific AI assistants.

Why it matters: Scientific AI tools are being deployed for literature review and data extraction; a model that fabricates values for blurred axes or trusts wrong captions poses real risks.

Method: 250 expert-annotated figures extended via image transformations, resistance probes, caption-bias probes, and selective-blur targets, producing 34,000+ evaluation setups.

Result: VLMs confabulate on 62–84% of blur conditions; caption-bias rate 41–67%; standard accuracy rank correlates poorly with behavioral reliability ($\rho = 0.31$).

Evidence: Resistance probe failures (71% compliance with false premises) confirm that models do not systematically detect their own visual limitations.

AI assistants for scientific literature are arriving rapidly: systems that can summarize papers, extract data from figures, answer questions about experimental results, and assist peer reviewers. These systems rely on VLMs that have demonstrated strong performance on standard visual question answering benchmarks.

Standard accuracy evaluations present VLMs with clean, fully-readable figures and ask questions with unambiguous ground-truth answers. This measures one important capability — accurate perception and reasoning — but ignores a second capability that is equally important for scientific use: *knowing what you cannot see*.

When a research figure has a blurred axis label, a partially occluded data series, or a misleading caption inserted during an adversarial or unintentional editing step, a reliable scientific AI assistant should express appropriate uncertainty. Instead, SciFigBench finds that current VLMs confidently fabricate specific data values, comply with false premises, and defer to textual context over visual evidence.

Why Existing Approaches Fall Short

Standard VQA and figure understanding benchmarks evaluate on clean images with unambiguous ground truth. They do not assess:

- Whether models appropriately express uncertainty when evidence is missing (rather than guessing)
- Whether models are biased by captions that contradict visual content
- Whether models detect fine-grained degradation of specific visual elements (e.g., one axis blurred while another is intact)

Hallucination benchmarks typically test factual grounding in natural images (“is there a cat in the image?”) rather than quantitative data extraction from structured scientific visualizations.

SciFigBench fills the gap by systematically varying the *state* of the visual evidence rather than just the questions asked about it.

Core Mechanism

Base figures: 250 scientific figures collected from STEM research papers across biology, physics, chemistry, computer science, and medicine. Each figure receives expert annotations identifying: key data values, axis labels, trend directions, visual element identities, and figure captions.

Evaluation setups (>34,000 total):

- (1) **Image transformations:** Gaussian blur, label removal, color-region masking applied to specific ele-

The Big Picture

ments while leaving others intact.

- (2) **Reasoning questions:** Quantitative (“What is the value at $x = 3$?”) and qualitative (“Which series shows the steeper slope?”).
- (3) **Resistance probes:** Model is told (falsely) that a specific value can be read from a blurred region — tests whether it complies or refuses.
- (4) **Caption-bias probes:** Figure paired with a caption that states a trend opposite to what the graph shows — tests whether model trusts visual evidence or caption.
- (5) **Selective-blur targets:** Single data elements blurred while the rest of the figure is intact — tests fine-grained awareness of local information availability.

Technical Formulation

The **behavioral reliability score (BRS)** for model M on perturbation condition set $\mathcal{P}$:

$$\text{BRS}(M) = \frac{1}{|\mathcal{P}|} \sum_{c \in \mathcal{P}} \frac{1}{|Q_c|} \sum_{q \in Q_c} \mathbf{1}[M(f_c, q) \text{ is behaviorally appropriate}]$$

where f_c is figure f under perturbation c , Q_c is the question set for condition c , and “behaviorally appropriate” means the model expresses uncertainty or refuses when evidence is unavailable, and trusts visual evidence over conflicting captions.

The **caption-bias score (CBS)**:

$$\text{CBS}(M) = \mathbb{E}_{(f,q)} [\mathbf{1}[M(f, q, \text{cap}_{\text{wrong}}) \neq M(f, q, \text{no cap})]]$$

where $\text{cap}_{\text{wrong}}$ is a caption that contradicts visual truth.

The **refusal rate (RR)** on resistance probes:

$$\text{RR}(M) = \mathbb{E}_{(f,q,\text{probe})} [\mathbf{1}[M \text{ declines to provide value}]]$$

Learning or Inference Procedure

SciFigBench is an evaluation benchmark with no training component. Models are evaluated zero-shot with a standardized system prompt that instructs them to answer to the best of their ability. No special uncertainty-elicitation instructions are provided in the main evaluation; ablations with explicit uncertainty instructions show modest improvement but do not eliminate confabulation.

What the Guarantee Says

No formal guarantee is provided. Ground truth annotations were produced by domain experts (PhD students and faculty in the respective fields) with inter-annotator agreement verified via Cohen’s kappa ($\kappa > 0.82$ across all annotation types). Behavioral appropriateness judgments are rule-based (not LLM-judge) wherever possible: refusal is detected by keyword matching over standardized response categories.

Experimental Findings

- **Confabulation under blur:** VLMs provide specific numerical answers 62–84% of the time when the asked-about visual element is blurred, nearly unchanged from clean conditions.
- **Caption bias:** Models align with the contradictory caption over correct visual evidence 41–67% of the time. Larger models show more bias, not less.
- **Resistance probe compliance:** Models comply with false premises (generate values from blurred regions) 71% of the time. Refusal rate is below 15% for all evaluated models.
- **Rank-reliability correlation:** Spearman $\rho = 0.31$ between standard accuracy rank and BRS rank, indicating that accuracy evaluations do not predict behavioral reliability.
- **Domain variation:** Biology and chemistry figures show higher caption-bias rates than computer science figures, possibly due to differences in training data figure-caption co-occurrence.

Ablations and Interpretation

Uncertainty elicitation: Prompting models to “express uncertainty when you cannot read a value” reduces confabulation from 74% to 52% — improvement, but not elimination. Models still confabulate more than half the time despite explicit instruction.

Blur severity: Confabulation rate is surprisingly stable across blur intensities from mild ($\sigma = 2$) to severe ($\sigma = 20$). This suggests models are generating values from semantic expectations rather than partially visible pixel information.

Caption-bias by model size: The largest evaluated models (7B–70B parameters) exhibit more caption bias than smaller models (1B–3B), consistent with larger models placing more weight on linguistic context relative to visual evidence during generation.

Selective vs. global blur: When only a single axis label is blurred while the rest of the figure is intact, confabulation rate (78%) slightly exceeds the global-blur rate (74%), suggesting models do not notice fine-grained local information loss.

Reference

Paul Osemudiamé Oamen, Owusu-Banahene Osei, Ananya Mukherjee, Christian Greisinger, Steffen Eger, Pius Onobhayedo, Wei Zhao. **How Do VLMs Behave When Blind or Misled? Behavioral Evaluation of VLMs on Scientific Figures.** arXiv:2608.13267, August 2026. <https://arxiv.org/abs/2608.13267>